AMAN KUMAR SINGH

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SUMMARY

Electrical & Electronics Engineering student specializing in embedded cybersecurity and hardware-enforced fail-closed architectures. Inventor of four drafted patent applications spanning battery security, medical-device anti-spoofing, earthen primary energy, and spacecraft radiation fault-tolerance. 2,150+ GitHub commits and a 217-day streak across defense-tech DSP, on-device AI, and deployed commercial platforms.

PATENT APPLICATIONS (DRAFTED — FILING IN PROGRESS)

BioTrustPulse — Fail-Closed Hardware-Interlocked Biometric Diagnostic Apparatus
A fail-closed medical-diagnostic apparatus that verifies device integrity before any measurement, preventing sensor spoofing and tampering.

Sentry BMS — Cyber-Physical Battery Management & Trust-Gated Access Control System
A cyber-physical battery-management and access-control system that fuses battery-health integrity with user authentication: tampering with the battery automatically cuts power and denies physical access.

Agastya — Earthen Primary Galvanic Power Generation System
A battery-free, earth-abundant primary power system delivering a stable regulated supply for off-grid loads and on-demand hydrogen generation.

RadZero Space AI — Spacecraft Attitude Control & Radiation Fault-Mitigation Apparatus
A spacecraft attitude-control apparatus that detects and repairs radiation-induced memory faults in on-board AI in real time, maintaining stable control without a system reboot.

EXPERIENCE

Founder — Fauji Niwas & ScrapMandi | React, Flutter, Firebase, Razorpay | Feb 2026 – Present

- Built and deployed two production platforms: a privacy-first defense relocation PWA for military personnel (AES-256-GCM + RSA-OAEP, zero-PII server exposure) and a B2B scrap-exchange marketplace (React 18 + Firebase, 7,000+ lines, server-side Razorpay escrow verification).

PROJECTS

NIRDHVANI — Tactical Throat-Acoustic ANC System (SIH) | ESP32, TinyML, DSP | Aug 2026

- Throat-contact piezo ANC for 120 dB+ battlefield noise; NLMS adaptive filter + TinyML step-size tuning, <4 ms latency, prototype BOM under Rs 780.

VARUNA-SDS — Subsea Software-Defined Sonar Payload (SIH) | STM32, DMA/DAC | Aug 2026

- 10–100 kHz LFM chirp synthesis with 7.6 cm range resolution (50x over fixed-frequency sonar); Tukey/Hann windowing for multipath environments.

Eloqui — 100% Offline AI Speaking Coach | Flutter, C++ FFI, llama.cpp | Jul 2026

- On-device LLM inference + Whisper STT + Piper TTS; solved the multi-model RAM cliff via deterministic lifecycle management (peak RAM < 1.2 GB).

TECHNICAL SKILLS

Hardware: C/C++, FreeRTOS, ESP32, STM32, DSP (NLMS/LFM), TinyML, DMA/ADC/DAC/PWM, PCB & SMD soldering
Security: AES-256-GCM, RSA-OAEP, ECDSA/ECDH P-256, ATECC608A secure element, Secure Boot v2, STRIDE, fail-closed interlocks
Software: Python, React 18, TypeScript, Flutter/Dart, Firebase, Razorpay API, Git, Linux

EDUCATION

Galgotias College of Engineering & Technology (AKTU) | B.Tech, Electrical & Electronics Engineering | 3rd Semester | Expected 2029
LBS Convent School, Kota (CBSE Class XII, PCM) | 72.4% | JEE Main Mathematics: 90+ percentile

CERTIFICATIONS & ACHIEVEMENTS

NCC Air Wing 'A' Certificate (Ministry of Defence) | Duolingo English Test 130/160 (CEFR C1)
Cisco: Introduction to Cybersecurity | Deloitte Cyber Job Simulation (Forage) | The Future of Chiplets (EE Times / Goldcast)
GeeksforGeeks: Computer Fundamentals | MathWorks: MATLAB Onramp & Core MATLAB Skills | ULSA Academic Honor